

Essential Logical Equivalences Reference

Identity Laws

Law	Equivalence	Explanation
Identity for $\wedge$	$p \wedge T \equiv p$	Conjunction with tautology leaves p unchanged
Identity for $\vee$	$p \vee F \equiv p$	Disjunction with contradiction leaves p unchanged

Domination Laws

Law	Equivalence	Explanation
Domination for $\wedge$	$p \wedge F \equiv F$	Conjunction with false is always false
Domination for $\vee$	$p \vee T \equiv T$	Disjunction with true is always true

Idempotent Laws

Law	Equivalence	Explanation
Idempotent for $\wedge$	$p \wedge p \equiv p$	Repeating conjunction doesn't change truth
Idempotent for $\vee$	$p \vee p \equiv p$	Repeating disjunction doesn't change truth

Double Negation Law

Law	Equivalence	Explanation
Double Negation	$\neg(\neg p) \equiv p$	Two negations cancel out

Commutative Laws

Law	Equivalence	Explanation
Commutative for $\wedge$	$p \wedge q \equiv q \wedge p$	Order doesn't matter in conjunction
Commutative for $\vee$	$p \vee q \equiv q \vee p$	Order doesn't matter in disjunction

Associative Laws

Law	Equivalence	Explanation
Associative for $\wedge$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Grouping doesn't matter in conjunction
Associative for $\vee$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	Grouping doesn't matter in disjunction

Distributive Laws

Law	Equivalence	Explanation
$\vee$ distributes over $\wedge$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	Disjunction distributes over conjunction
$\wedge$ distributes over $\vee$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Conjunction distributes over disjunction

De Morgan's Laws (CRITICAL)

Law	Equivalence	Explanation
De Morgan for $\wedge$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$	Negate AND $\rightarrow$ OR of negations
De Morgan for $\vee$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	Negate OR $\rightarrow$ AND of negations

Memory tip: Negation flips the operator ($\wedge \leftrightarrow \vee$) and negates each component.

Absorption Laws

Law	Equivalence	Explanation
Absorption 1	$p \vee (p \wedge q) \equiv p$	p absorbs $p \wedge q$
Absorption 2	$p \wedge (p \vee q) \equiv p$	p absorbs $p \vee q$

Negation Laws

Law	Equivalence	Explanation
Negation for $\vee$	$p \vee \neg p \equiv T$	Always true (tautology)
Negation for $\wedge$	$p \wedge \neg p \equiv F$	Always false (contradiction)

Conditional Equivalences ★ (CRITICAL)

Law	Equivalence	Explanation
Conditional as Disjunction	$p \rightarrow q \equiv \neg p \vee q$	Most important for eliminating conditionals
Contrapositive	$p \rightarrow q \equiv \neg q \rightarrow \neg p$	Essential for proof by contrapositive
Negation of Conditional	$\neg(p \rightarrow q) \equiv p \wedge \neg q$	Conditional is false only when p is true and q is false

Biconditional Equivalences ★

Law	Equivalence	Explanation
Biconditional Definition	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$	Both directions must hold
Biconditional as Conjunction/Disjunction	$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$	True when both have same truth value

Law	Equivalence	Explanation
Biconditional as Negated XOR	$p \leftrightarrow q \equiv \neg(p \oplus q)$	Opposite of exclusive or
Negation of Biconditional	$\neg(p \leftrightarrow q) \equiv p \oplus q$	Equivalent to exclusive or

Additional Conditional Forms

English Form	Logical Form	Notes
If p, then q	$p \rightarrow q$	Standard conditional
p only if q	$p \rightarrow q$	"Only if" introduces conclusion
p implies q	$p \rightarrow q$	Direct statement
q if p	$p \rightarrow q$	Reversed word order
q whenever p	$p \rightarrow q$	Alternative phrasing
p is sufficient for q	$p \rightarrow q$	p guarantees q
q is necessary for p	$p \rightarrow q$	p requires q
q unless $\neg p$	$p \rightarrow q$	"Unless" means "if not"

Quantifier Negations (For Later Topics)

Statement	Negation	Explanation
$\forall x P(x)$	$\exists x \neg P(x)$	Not all $\rightarrow$ at least one counterexample
$\exists x P(x)$	$\forall x \neg P(x)$	Not exists $\rightarrow$ all fail the property

Common Proof Strategies Using These Equivalences

1. Simplification:

- Use identity and domination laws to eliminate T and F
- Apply idempotent laws to remove duplicates
- Use absorption to simplify nested expressions

2. Converting Conditionals:

- Replace $p \rightarrow q$ with $\neg p \vee q$ to eliminate conditionals
- Use contrapositive when the negated form is easier to work with

3. Negating Complex Statements:

- Apply De Morgan's laws systematically
- Remember: $\neg(p \rightarrow q) \equiv p \wedge \neg q$
- For quantifiers: flip the quantifier and negate the predicate

4. Proving Equivalences:

- Chain multiple equivalences together
- Work from one side toward the other
- Or work from both sides toward a common middle form

Quick Reference: Most Used in Proofs

1. **Conditional as Disjunction:** $p \rightarrow q \equiv \neg p \vee q$
2. **Contrapositive:** $p \rightarrow q \equiv \neg q \rightarrow \neg p$
3. **De Morgan's Laws:** $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$
4. **Negation of Conditional:** $\neg(p \rightarrow q) \equiv p \wedge \neg q$
5. **Double Negation:** $\neg(\neg p) \equiv p$
6. **Distributive Laws:** Both forms